

DAY 2

1. List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply

SMPS (Switched Mode Power Supply)

Laptop charger – Uses an SMPS to efficiently convert AC mains power into the low-voltage DC needed to charge the laptop.

Desktop computer (CPU power supply) – Uses an SMPS to provide stable DC voltages (such as 12 V, 5 V, and 3.3 V) to internal components with high efficiency.

LED TV – Uses an SMPS to convert AC power into regulated DC voltages while reducing power loss and heat.

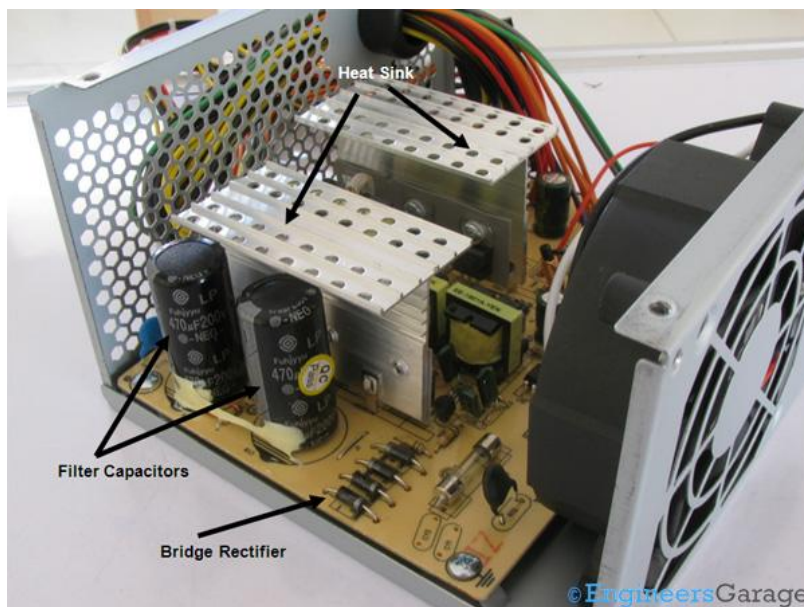
UPS (Uninterruptible Power Supply)

Desktop computer with UPS – A UPS provides instant backup power during outages, preventing sudden shutdowns and data loss.

Wi-Fi router connected to a UPS – A UPS keeps the router running during power cuts, maintaining internet connectivity.

2. Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections You can hand-draw and upload a photo, or use any free diagram tool like draw.io

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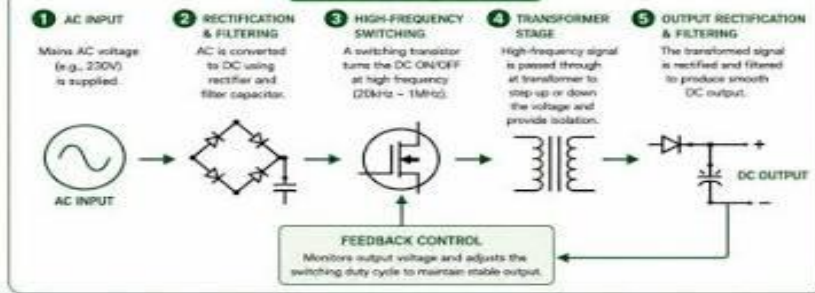
SMPS

SWITCHED MODE POWER SUPPLY

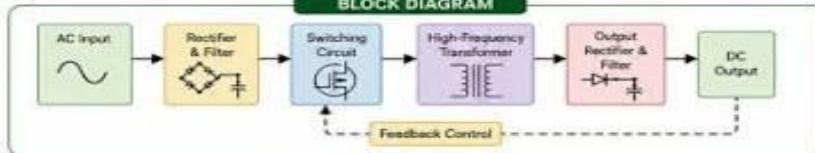
SMPS is an electronic power supply that converts electrical power efficiently using high-frequency switching devices.



WORKING PRINCIPLE



BLOCK DIAGRAM



ADVANTAGES



An SMPS (Switch Mode Power Supply) converts standard AC mains into stable, low-voltage DC for your PC. It operates through four key stages: the AC Input filters noise, the Primary Rectifier converts AC to DC, the Transformer/Switching chops and steps it down, and the DC Output smooths it for use.

1. AC Input Section (Primary Side)

- **AC Mains Input:** Receives $\sim 230\text{V}$ or $\sim 115\text{V}$ AC power from the wall socket.
- **EMI/RFI Filter:** A network of inductors and capacitors that stops electromagnetic interference generated by the power supply from leaking back into the mains and corrupting other devices.
- **Fuse:** Protects the circuit by blowing if there is an extreme current overload or short circuit.

2. Rectifier & Primary Filter Section

- **Bridge Rectifier:** A configuration of diodes that converts the incoming AC voltage into a high-voltage, pulsating DC voltage.
- **Primary Filter Capacitors:** High-voltage capacitors that smooth the pulsating DC into a relatively steady but unregulated DC (often around $\sim 320\text{V}$ to $\sim 340\text{V}$).

3. High-Frequency Switching & Transformer Section

- **Switching Transistor (MOSFET/IGBT):** Acts as a high-speed electronic switch that turns the high-voltage DC on and off very rapidly (typically $\sim 20 \text{ kHz}$ to $\sim 1 \text{ MHz}$) to create a high-frequency AC square wave.
- **PWM (Pulse Width Modulation) Controller:** An IC that monitors the output voltage and adjusts the duty cycle of the switching transistor, dictating exactly how much power passes through.
- **Main Transformer:** Steps down the high-frequency/high-voltage AC signal to the lower voltages needed by the PC (e.g., $\sim 12 \text{ V}$, $\sim 5 \text{ V}$, $\sim 3.3 \text{ V}$). Because the switching frequency is very high, this transformer can be much smaller than traditional linear transformers.

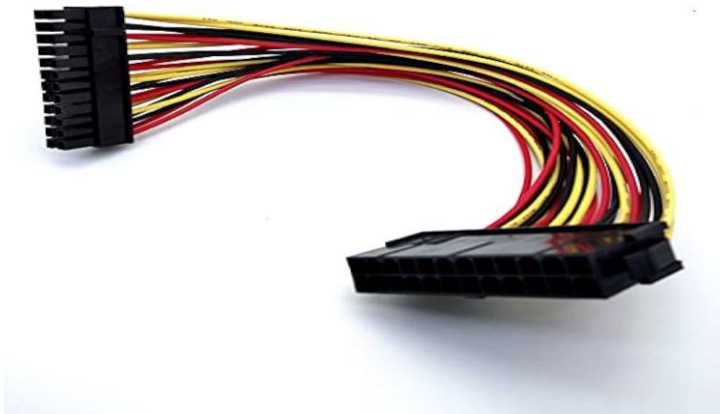
4. Output Section (Secondary Side)

- **Output Rectifiers:** Fast-switching Schottky diodes (or secondary MOSFETs) that convert the stepped-down high-frequency AC back into direct current (DC).
- **Output Filters (Inductors & Capacitors):** Chokes (inductors) and low-voltage capacitors remove final voltage "ripples" to deliver smooth, stable, and highly regulated DC to the PC components.

3. Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.

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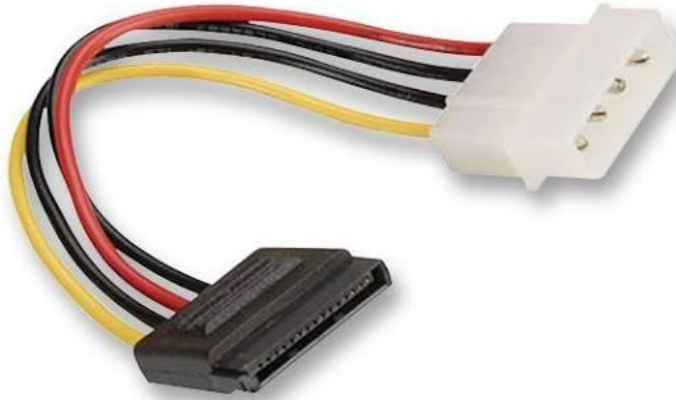
ATX 24-pin Motherboard Connector



Supplies power to the motherboard.

Turn off the PC and unplug the power cable before connecting or disconnecting it.

SATA Power Connector



Powers SATA hard drives (HDDs), SSDs, and DVD/Blu-ray drives.
Align the L-shaped connector correctly and do not force it into the port.

Molex (4-pin) Connector

Powers older hard drives, optical drives, cooling fans, and some accessories.

Hold the connector by its plastic housing when unplugging; do not pull on the wires.



PCIe 6-pin / 8-pin Power Connector



Supplies extra power to graphics cards (GPUs).

Make sure the connector clicks into place before turning on the computer.

4. Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.

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Steps to Connect an SMPS in a New Gaming PC

1. **Turn off the power supply** and unplug the power cable before starting the installation.
2. **Connect the motherboard power** using the **24-pin ATX power connector** from the SMPS to the motherboard's 24-pin power socket.
3. **Connect the CPU power** using the **4+4-pin (8-pin) EPS/CPU power connector** to the CPU power socket near the processor.
4. **Connect the graphics card (GPU)** using the required **PCIe 6-pin or 8-pin power connector(s)** from the SMPS to the graphics card.

5. **Connect SATA storage devices** (SSD, HDD, or DVD drive) using the **SATA power connector** from the SMPS.
6. **Connect any older devices or case fans**, if required, using the **4-pin Molex connector**.
7. **Check all connections** to ensure every connector is fully inserted and locked into place.
8. **Connect the AC power cable** to the SMPS, switch on the power supply, and power on the PC to verify that all components are working correctly.